Unit: 6 Unit 8: Quadratics
Topic: Converting Forms (Completing the Square)

Name: ________________
Date: ___________

Quiz (Carolina Reaper)

Q1. A shipping company's cost C to transport x containers is given by $C = \mathrm{x}^2 + 10x + 30$. Convert this equation to vertex form:
(A) $C = (\mathrm{x}+5)^2 + 5$
(B) $C = (\mathrm{x}-5)^2 - 5$
(C) $C = (\mathrm{x}+5)^2 - 5$
(D) $C = (\mathrm{x}-5)^2 + 5$
(E) $C = \mathrm{x}^2 + 10x + 30$

Q2. The traffic flow T through a tunnel is modeled by $T = -2\mathrm{x}^2 - 12x - 18$, where x is the number of lanes closed. Complete the square to rewrite T in vertex form:
(A) $T = -2(\mathrm{x}+3)^2 - 0$
(B) $T = -2(\mathrm{x}-3)^2 + 0$
(C) $T = -2(\mathrm{x}+3)^2 + 0$
(D) $T = -2(\mathrm{x}-3)^2 - 0$
(E) $T = -2\mathrm{x}^2 - 12x - 18$

Q3. The time t it takes for a delivery truck to travel x miles is given by $t = \mathrm{x}^2 + 4x + 8$. Identify the vertex of the parabola:
(A) (-2, 4)
(B) (2, 4)
(C) (-2, -4)
(D) (2, -4)
(E) (0, 8)

Q4. A logistics company's profit P from selling x units of a product is given by $P = -\mathrm{x}^2 + 12x + 20$. Complete the square to rewrite P in vertex form:
(A) $P = -(\mathrm{x}-6)^2 + 56$
(B) $P = -(\mathrm{x}+6)^2 - 56$
(C) $P = -(\mathrm{x}-6)^2 - 56$
(D) $P = -(\mathrm{x}+6)^2 + 56$
(E) $P = -\mathrm{x}^2 + 12x + 20$

Q5. The cost C of shipping x tons of cargo is given by $C = \mathrm{x}^2 - 8x + 12$. Convert this equation to vertex form:
(A) $C = (\mathrm{x}-4)^2 - 4$
(B) $C = (\mathrm{x}+4)^2 + 4$
(C) $C = (\mathrm{x}-4)^2 + 4$
(D) $C = (\mathrm{x}+4)^2 - 4$
(E) $C = \mathrm{x}^2 - 8x + 12$

Q6. The number of hours h it takes for a train to travel x miles is given by $h = \mathrm{x}^2 + 2x + 5$. Complete the square to rewrite h in vertex form:
(A) $h = (\mathrm{x}+1)^2 + 4$
(B) $h = (\mathrm{x}-1)^2 - 4$
(C) $h = (\mathrm{x}+1)^2 - 4$
(D) $h = (\mathrm{x}-1)^2 + 4$
(E) $h = \mathrm{x}^2 + 2x + 5$

Q7. A transportation company's revenue R from x trips is given by $R = -2\mathrm{x}^2 + 20x + 10$. Identify the vertex of the parabola:
(A) (5, 60)
(B) (-5, 60)
(C) (5, -60)
(D) (-5, -60)
(E) (0, 10)

Q8. The time t it takes for a plane to fly x miles is given by $t = \mathrm{x}^2 - 6x + 10$. Complete the square to rewrite t in vertex form:
(A) $t = (\mathrm{x}-3)^2 + 1$
(B) $t = (\mathrm{x}+3)^2 - 1$
(C) $t = (\mathrm{x}-3)^2 - 1$
(D) $t = (\mathrm{x}+3)^2 + 1$
(E) $t = \mathrm{x}^2 - 6x + 10$

Open Ended Questions (FRQ)

Q9. A delivery company's cost C to transport x packages is given by $C = \mathrm{x}^2 + 8x + 15$. Convert this equation to vertex form and identify the vertex of the parabola. Show all work and explain your reasoning.

Q10. The traffic flow T through a tunnel is modeled by $T = -3\mathrm{x}^2 - 18x - 12$, where x is the number of lanes closed. Complete the square to rewrite T in vertex form and identify the vertex of the parabola. Show all work and explain your reasoning.

Chef's Tips

- Read each question carefully
- Use the completing the square method when necessary
- Show all work and explain reasoning for free response questions